COL703/7203

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$\neg(p \to q) \vdash q \to p$

- q — assumption
  - p — assumption
  - q — copy
- $p \to q$
- $\bot$
- r
- p

$q \to p$

$\neg(p \to q) \vdash q \to p$

$\phi \vee \psi$ $\boxed{\begin{matrix}\phi \\ \chi\end{matrix}}$ $\boxed{\begin{matrix}\psi \\ \chi\end{matrix}}$

$q \vee \neg q$ LEM

q assumption

p assumption

q copy

$p \to q$

$\bot$ $\neg e$

$q \to p$

$\neg q$ assumption

q assumption

$\bot$

p

$q \to p$

χ

$q \to p$

$q \vee \neg q$

$q \qquad \qquad \neg q$

$p \vee \neg p \qquad \qquad p \vee \neg p$

$q \qquad q \qquad \neg q \qquad \neg q$

$\cancel{p} \qquad \cancel{\neg p} \qquad \cancel{p} \qquad \cancel{\neg p}$

Extra information (p or $\neg p$) should not hurt the proof.

Semantic Equivalence

$$\phi \equiv \psi \qquad \begin{array}{l} \phi \vDash \psi \\ \psi \vDash \phi \end{array}$$

The brute force way to prove this is through truth tables. One can, of course, appeal to soundness and completeness of prop. logic, and prove the validity of

$$\phi \vdash \psi \quad \text{and} \quad \psi \vdash \phi.$$

But we can also manipulate the expressions through known equivalences.

$$\phi \vee \phi \equiv \phi$$
$$\phi \wedge \phi \equiv \phi$$
Idempotence

$$\phi \wedge \psi \equiv \psi \wedge \phi$$
$$\phi \vee \psi \equiv \psi \vee \phi$$
Commutativity

$$(\phi_1 \wedge \phi_2) \wedge \phi_3 \equiv \phi_1 \wedge (\phi_2 \wedge \phi_3)$$
$$(\phi_1 \vee \phi_2) \vee \phi_3 \equiv \phi_1 \vee (\phi_2 \vee \phi_3)$$
Associativity

$$\phi \wedge (\phi \vee \psi) \equiv \phi$$
$$\phi \vee (\phi \wedge \psi) \equiv \phi$$
Absorption

$$\phi \wedge (\psi_1 \vee \psi_2) \equiv (\phi \wedge \psi_1) \vee (\phi \wedge \psi_2)$$

$$\phi \vee (\psi_1 \wedge \psi_2) \equiv (\phi \vee \psi_1) \wedge (\phi \vee \psi_2)$$

Distributivity

$$\neg\neg\phi \equiv \phi \qquad \text{(Double negation)}$$

$$\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$$

$$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$$

De Morgan's Laws

$$\phi \vee \neg\phi \equiv true$$

$$\phi \wedge \neg\phi \equiv false$$

$$\phi \vee true \equiv true$$

$$\phi \vee false \equiv \phi$$

$$\phi \wedge true \equiv \phi$$

$$\phi \wedge false \equiv false$$

$$(\underline{\phi \vee (\psi_1 \vee \psi_2)}) \wedge (\psi_2 \vee \neg\phi)$$

$$\equiv \psi_2 \vee (\neg\phi \wedge \psi_1)$$

$$(\phi \vee \psi_1) \vee \psi_2$$

$$(\psi_2 \vee (\phi \vee \psi_1)) \wedge (\psi_2 \vee \neg\phi)$$

$$\underbrace{\psi_2}_{\alpha} \vee (\underbrace{(\phi \vee \psi_1)}_{\beta} \wedge \underbrace{\neg\phi}_{\gamma})$$

$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

$$\psi_2 \vee (\neg\phi \wedge (\phi \vee \psi_1))$$

$$\psi_2 \vee ((\neg\phi \wedge \phi) \vee (\neg\phi \wedge \psi_1))$$

Normal Forms (of propositional logic formulas)

Negation Normal Form (NNF)

A wff is in negation normal form if it only uses $\wedge$, $\vee$, and literals (atoms or their negation).

Every wff is equivalent to one in NNF.

Convert $\neg(p \rightarrow (p \wedge q))$ into NNF.

$$\neg(p \to (p \wedge q))$$

$$\neg(\neg p \vee (p \wedge q))$$

$$\neg\neg p \wedge \neg(p \wedge q)$$

$$= \quad p \wedge (\neg p \vee \neg q)$$

$$(p \wedge \neg p) \vee (p \wedge \neg q)$$

Disjunctive Normal Form

conjunctive

A wff is in disjunctive normal form if it is a disjunction conjunction of one or more terms where each term is a conjunction disjunction of one or more literals.

How do we derive these normal forms?

→ Manipulating the expression through known equivalences

→ Truth table:

p	q	r	ϕ
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$(\neg p \wedge \neg q \wedge \neg r)$
$\vee (\neg p \wedge q \wedge r)$
$\vee (p \wedge \neg q \wedge \neg r)$
$\vee (p \wedge q \wedge r)$

CNF
exercise

$$(\neg q \vee p \vee r) \wedge (\neg p \vee q \vee s) \wedge (r \vee \neg s \vee \neg q)$$

Claim: A disjunction of literals $(l_1 \vee l_2 \vee \dots \vee l_m)$ is valid iff there are $1 \le i, j \le m$ such that $l_i = \neg l_j$.

Proof: **Exercise**